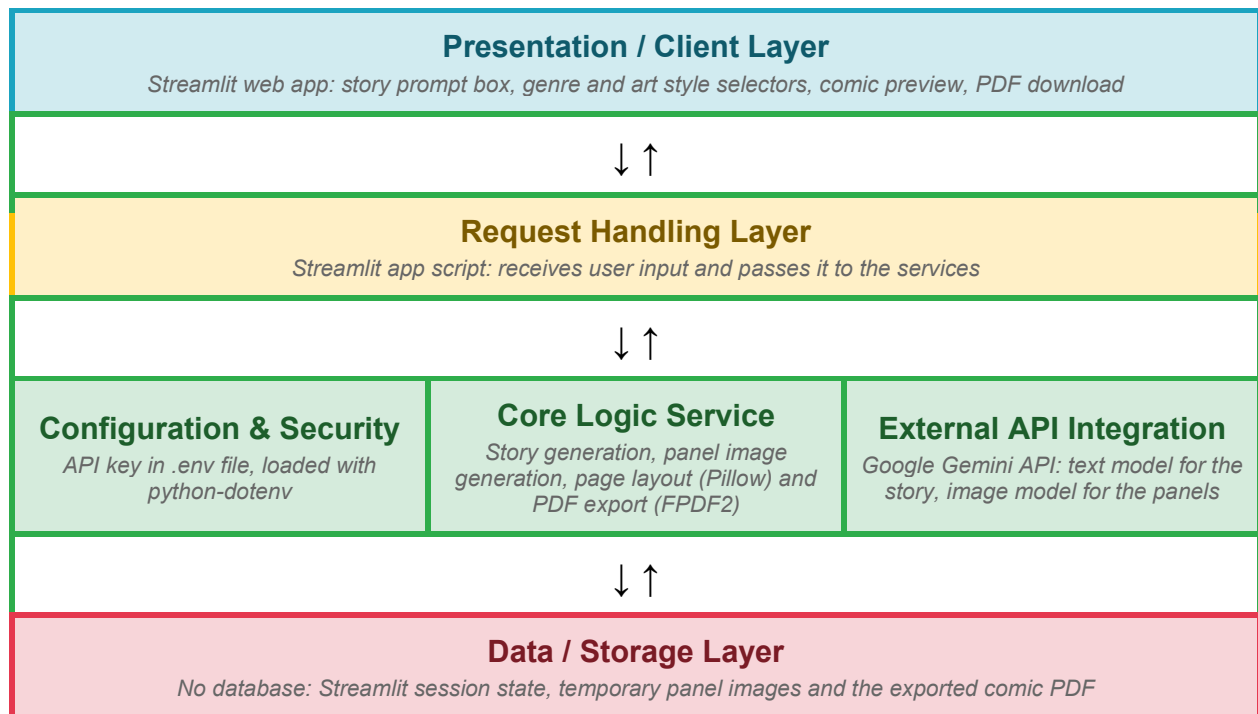


Solution Architecture

Date	1 October 2026
Team ID	6ab501466f290be8a6b4f5d3
Project Name	ComicCraft – AI Comic Story Creator
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Web interface where the user enters a story idea, picks a genre and art style, views the generated comic and downloads the PDF.	Streamlit (Python)
Request Handling Layer	Receives the user's input, checks that the story idea is not empty, and calls the story, image, layout and export steps in order.	Streamlit app script, Python
Core Logic Service	Builds the story script and panel prompts, arranges the panels with captions into comic pages, and exports the finished comic as a PDF.	Python, Pillow, FPDF2
External API Integration	Generates the story and panel script with the Gemini text model and creates the panel pictures with the Gemini image model.	Google Gemini API (google-genai)
Configuration & Security	Keeps the Gemini API key out of the source code and the GitHub repository.	python-dotenv, .env file
Data / Storage Layer	Holds the story, panel images and final PDF during the user's session. No database is used.	Streamlit session state, temporary files